

Apil Thapa

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Kathmandu, 03 44600, Nepal

EXPERIENCE

• DeepMind Creations

Junior AI Developer

Baneshwor, Kathmandu

- Built a recommendation system application and customer profiling based on past purchase activities.
- Developed a mass email generation system with the integration of LLMs and agents for email marketing.
- Created a sales pitch for salespersons with the integration of LLMs and text-to-speech models.
- Developed a full-stack scraping application using automation software and LLMs to generate leads from Google Maps.
- Built an LLM application for scraping content from multiple websites (e.g., emails, products).
- Developed a Retrieval-Augmented Generation (RAG) application for product inventory management.
- Performed sentiment analysis on scraped lead shops using customer reviews and strategized outreach using LLMs and various tools.
- Conducted product classification using LLM tools and agents.
- Developed a routing system using the Google Maps API for locating leads, providing directions, and optimizing traffic.
- Created a generative AI-based customer service chatbot application that queries from a database in real time.

EDUCATION

• Islington College

BSc (Hons) Computer Science - London Metropolitan University

2021 - 2025

Kathmandu, Nepal

- partnered with London Metropolitan University, UK

• New Horizon College

+2 in Science (Computer Stream)

2019 - 2021

Butwal, Nepal

- Grade: A (With Distinction)

COURSE TRAININGS

• Artificial Intelligence Training

Broadway Infosys

Jun-Sep 2023

• Python Django Training

Broadway Infosys

Dec 2023-Feb 2024

PROJECTS

• Generative AI & LLMs

Tools: LangGraph, FastAPI, Trainer API, LangChain, GROQ, Diffusers

- Developed Muktinath Bank chatbot (LangGraph+RAG)
- Contributed in Sanima Bank chatbot (FastAPI+real-time data)
- Fine-tuned LLAMA2, MISTRAL, GEMMA using Trainer API for clothing brand
- Built food recipe generator with LangChain+Django+Gemini API
- Developed PDF summarizer/QA system with RAG architecture
- Implemented QLORA/LORA quantization (GGUF) for CPU inference
- Code generation using GROQ API with Code LLAMA
- Integrated Stable Diffusion models for text-to-image
- Research/publications on fine-tuning methodologies
- Created quiz app with LangChain

• Employee Management System (WorkSphere)

Tools: Python, WebSockets, OpenCV, YOLO, NLP, LLMs

- Developed full-stack employee management platform with real-time features
- Implemented PIN-based employee registration with manager approval workflow
- Built real-time task management system with instant approval/rejection
- Created WebSocket-based chat system for employee-manager communication
- Designed anti-spoofing facial recognition attendance using OpenCV, YOLO
- Integrated sentiment analysis on employee feedback using NLP
- Developed knowledge-based chatbot system
- Implemented document query summarization using AI/LLMs
- Added social features: posts, comments, announcements with notifications

• Multimodal Emotion Recognition System

Tools: Wav2Vec2, DistilBERT, ViT, PyTorch

- Built audio-text-visual emotion recognition system
- Fine-tuned Wav2Vec2 for speech emotion analysis
- Implemented DistilBERT for text sentiment analysis
- Applied Vision Transformers for facial expression recognition
- Developed probabilistic fusion model for classification

• Drowsiness Detection System

Tools: YOLOv8, OpenCV, Python

- Developed real-time drowsiness detection using YOLOv8
- Created custom dataset for eye/facial drowsiness
- Trained model for closed-eye detection
- Implemented real-time video processing
- Designed for driver safety applications

• Finance Tracker Desktop App

Tools: .NET MAUI, Blazor, C#, SQLite

- Built cross-platform financial tracking app
- Implemented expense/income management
- Developed financial insights visualization
- Used SQLite for data storage
- Designed intuitive Blazor UI

• Byte Bazaar (E-commerce Website)

Tools: Java JEE, JSP, Servlets, MySQL

- Developed full-featured gadget e-commerce platform
- Implemented user auth, product catalog, shopping cart
- Built admin dashboard for inventory management
- Integrated secure payment processing

• License Plate Detection System

Tools: YOLOv8, Roboflow, Google Colab

- Created custom dataset with 500+ annotated images
- Trained YOLOv8 model (mAP: 0.92)
- Implemented real-time plate detection
- Developed performance visualization tools

• Django Web Applications

Tools: Django, Bootstrap, PostgreSQL

- Built To-Do App with CRUD operations

- Developed Blog App with user comments
- Created News Aggregator with API integration
- Designed responsive frontend interfaces

• **RFID Security Gate System**

Tools: YOLOv4, RFID, Python, Streamlit

- Integrated RFID auth with vehicle detection
- Developed real-time gate control system
- Built monitoring dashboard with Streamlit
- Achieved 95% detection accuracy

• **Cold Mail Generator**

Tools: Groq, LangChain, Streamlit

- Automated personalized cold email generation
- Scraped/extracted job listing data
- Matched requirements with portfolio links
- Generated context-aware email templates

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, JavaScript
- **Frontend Development:** HTML5, CSS3, Bootstrap, Figma, Photoshop
- **Backend Frameworks:** Django, Flask, FastAPI, Streamlit, J2EE
- **Database Systems:** MySQL, Oracle SQL, SQLite
- **Data Science & ML:** Scikit-learn, TensorFlow, Keras, PyTorch, LLMs, LangChain, Langgraph
- **Data Processing & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, Power BI
- **Computer Vision:** OpenCV, YOLO, MediaPipe
- **NLP & AI Tools:** Transformers, NLTK, spaCy, Ollama, Groq
- **Mathematics for ML:** Linear Algebra, Probability, Statistics, Calculus, Matrices
- **DevOps & Deployment:** Git

CERTIFICATIONS

* AWS Certified <i>Machine Learning Foundations Amazon Web Services</i> <i>Machine Learning for NLP Amazon Web Services</i>	<i>Nov 2023</i>
* AWS Cloud Foundations <i>Amazon Web Services</i>	<i>Oct 2023</i>
* Deep Learning & Gen AI Specialization <i>DeepLearning.AI</i>	<i>Dec 2022</i>

REFERENCES

Name:	Aayush Adhikari
Title:	CTO Lead Data Scientist AI Consultant
Company:	DeepMind Creations
Contact:	aceraayush@gmail.com LinkedIn Profile
Relation:	AI Mentor & Project Supervisor
Endorsement:	<i>"Apil demonstrates exceptional proficiency in AI/ML, with strong capabilities in deep learning architectures and practical implementation of computer vision and NLP solutions."</i>